

FACULTY OF ENGINEERING AND BASIC SCIENCES
ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE

COURSE: ETL

Workshop-003: ETL Process using Apache Kafka + Machine Learning

1. Introduction

In this workshop, you will integrate Machine Learning with Data Streaming to develop a complete predictive system for happiness scores across different countries and years.

You are provided with five CSV files (data.zip), each containing data on happiness scores for different countries in various years. Your task is to design an end-to-end workflow that includes Exploratory Data Analysis (EDA), feature engineering, model training, data streaming, and prediction storage.

The overall goal is to train a regression model capable of predicting the *Happiness Score* using relevant features extracted from the provided datasets, according to EDA. Once the model is trained, you will use Kafka to stream the transformed data (features) and process it through a consumer that loads the trained model to generate predictions. The predictions and their corresponding input features will then be stored in a database for analysis and reporting.

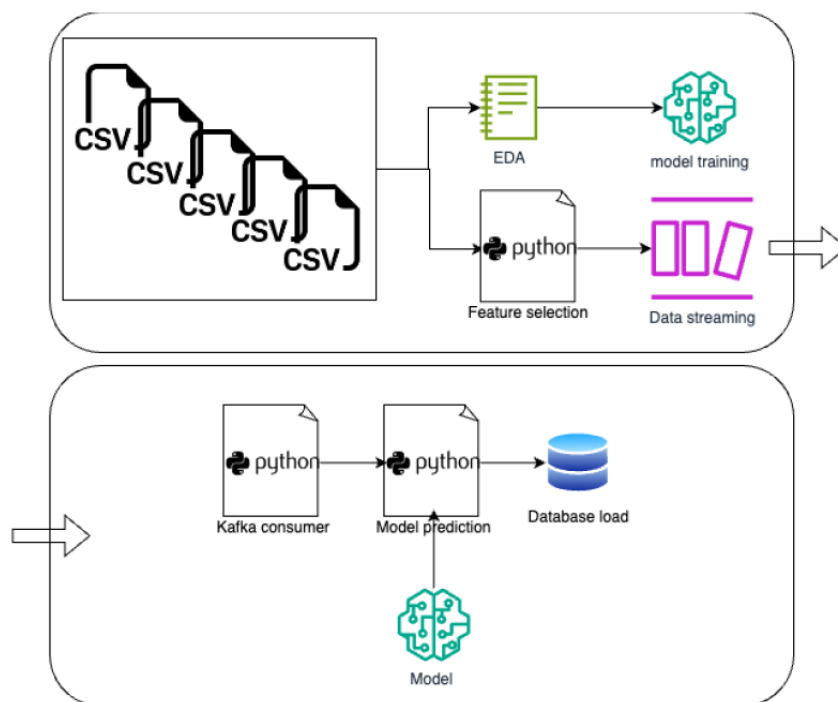


Fig. 1: Block Diagram

2. Learning Objectives

By completing this workshop, you will be able to:

1. Conduct EDA to understand and compare datasets from multiple years.

2. Perform ETL (Extract, Transform, Load) processes to clean and integrate heterogeneous data sources.
3. Select and engineering features suitable for regression modeling.
4. Train and evaluate a regression machine learning model with a train-test split (70%-30%).
5. Implement a Kafka-based system to process transformed data in real time.
6. Use a trained model (serialized as .pkl) to make predictions on streaming data.
7. Store input features and predicted values in a database for later analysis.
8. Compute and interpret performance metrics for the regression model.

3. Task Description

1. Data Preparation and EDA

- Load and inspect the five CSV files corresponding to different years of happiness data.
- Identify common and unique columns across datasets.
- Handle missing values, outliers, and inconsistent data formats.
- Generate descriptive statistics and visualizations to explore relationships among variables (e.g., GDP, social support, life expectancy).
- Select and preprocess the features that are most relevant for predicting the happiness score.

2. Model Training

- Split the prepared dataset into:
 - 70% training data
 - 30% testing data
- Train a regression model (e.g., Linear Regression).
- Save the trained model as a .pkl file for later use.

3. Data Streaming with Kafka

- Implement a Kafka Producer that reads the transformed data (features) and streams it record by record.
- Implement a Kafka Consumer that:
 - Receives the streamed data.
 - Loads the trained model (.pkl file).
 - Predicts the happiness score for each incoming record.
 - Stores the input features, the actual output, and the predicted score in a database (e.g., PostgreSQL, MySQL, or SQLite).

4. Model Evaluation and Reporting

- Compare the predicted scores against the actual testing data.
- Compute a performance metric (e.g., Mean Absolute Error or R^2 score).
- Summarize findings from the EDA and model evaluation.

5. Deliverables in GitHub

- a. README.md explaining your project, setup instructions, and key decisions.
- b. EDA Notebook:

- A Jupyter Notebook or Python script showing data loading, cleaning, visualization, and feature selection.
- c. Training Notebook and Trained Model File (.pkl):
 - A Jupyter Notebook or Python script showing model training using selected features.
 - The serialized model used for predictions in the streaming system.
- d. Kafka Producer and Consumer Code:
 - Complete and executable scripts for streaming and consuming data.
- e. Database with Predictions:
 - A database (or exported table) containing:
 - o Input features
 - o Actual happiness scores
 - o predicted happiness scores
- f. Performance
 - Visualizations and KPIs that show the performance of the model.
- g. Summary report:
 - Description of the datasets
 - Key findings from EDA
 - Model selection and training process
 - Evaluation metrics
 - Discussion of the streaming process and results

4. Technologies

You should use:

- Python
- Database (you choose)
- Apache Kafka
- CSV files
- Visualization tool.
- Scikit-learn library for machine learning

5. Evaluation Criteria

Item	Description	Weight
GitHub Repo	Repo created, organized, clean structure	0.5
EDA/Analysis	EDA to understand the data	0.5
Feature selection	Select and preprocess the features that are most relevant for predicting the happiness score	0.5
Kafka Producer	Implement a Kafka Producer that reads the transformed data (features) and streams it record by record.	0.5
Kafka consuming	Implement a Kafka Consumer that receives the streamed data.	0.5
Model training	Train a regression model	0.5
Model load/prediction	Loads the trained model (.pkl file) and predicts the happiness score for each incoming record.	0.5
KPIs & Visualizations	Correct metrics, accurate, clear charts	0.5
Documentation	Explanation of ETL pipeline, challenges, assumptions	1.0